

# Network Layer (1)

COMP90007 Internet Technologies

Lecturer: Ling Luo

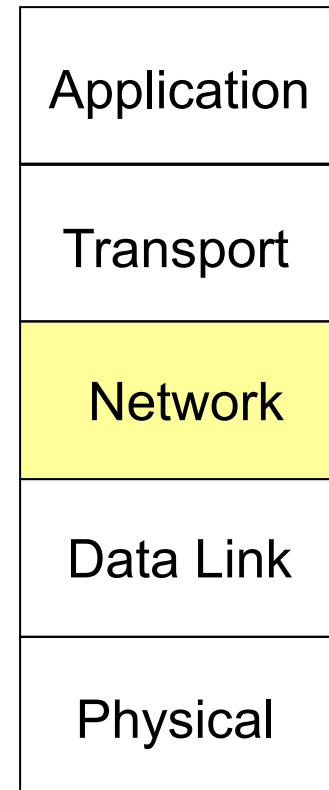
Semester 1, 2026

# Network Layer

**Connecting different networks  
(internetworking), routing**

Framing, error and flow control,  
medium access control

Different transmission media, signals,  
modulation ...

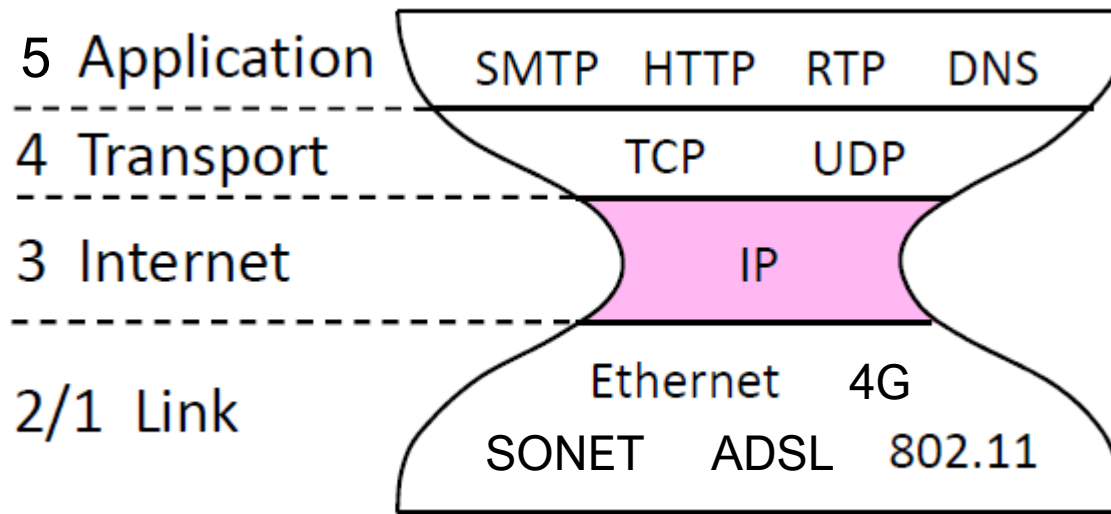


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# Outline

- Network layer on the Internet
- Types of services
- Internetworking
  - Tunneling
  - Fragmentation
  - Path MTU discovery
- Internet Protocol
  - Addressing
  - Subnetting
- Routing algorithms

# Network Layer on the Internet (1)



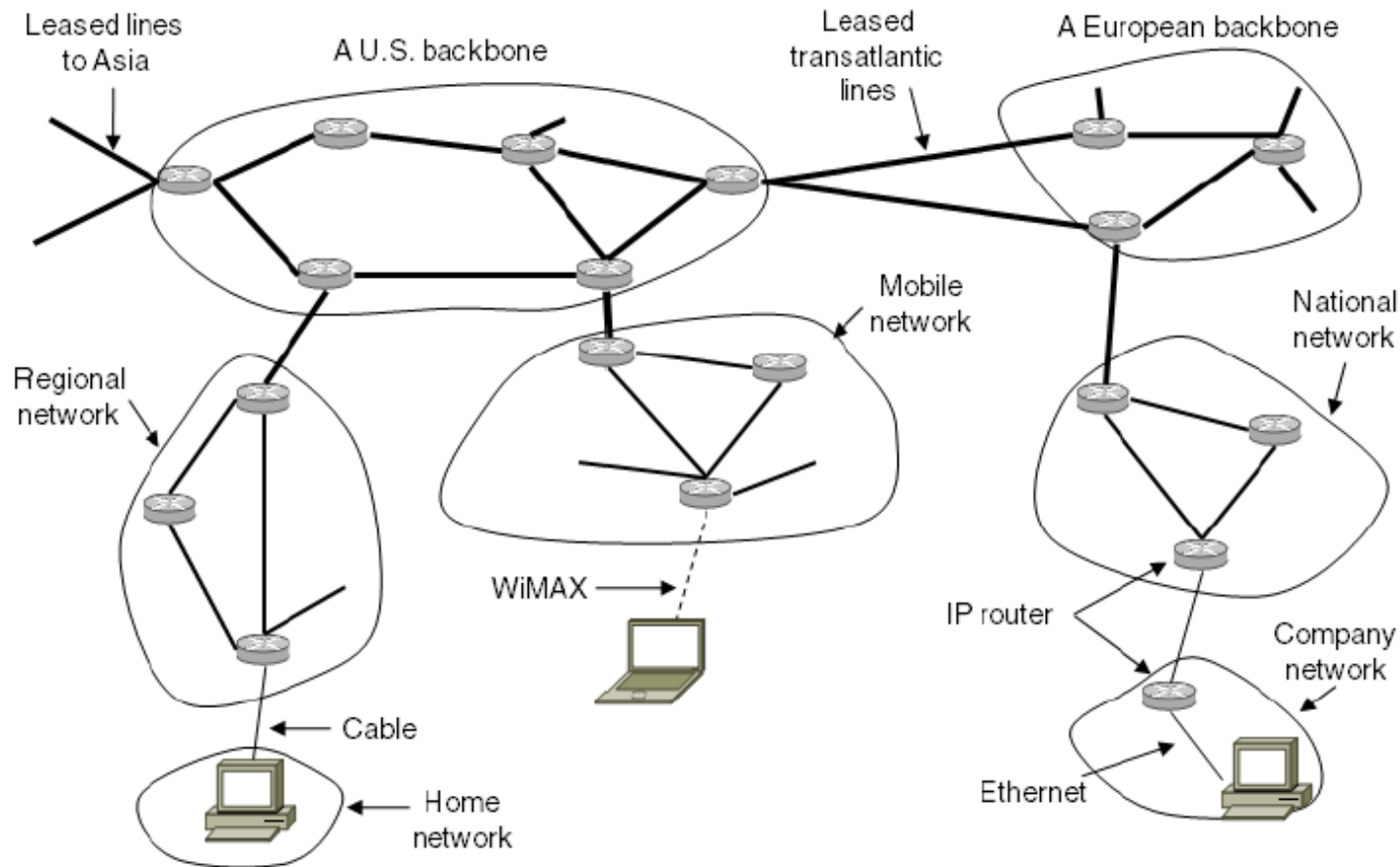
**TCP/IP model**

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# Network Layer on the Internet (2)

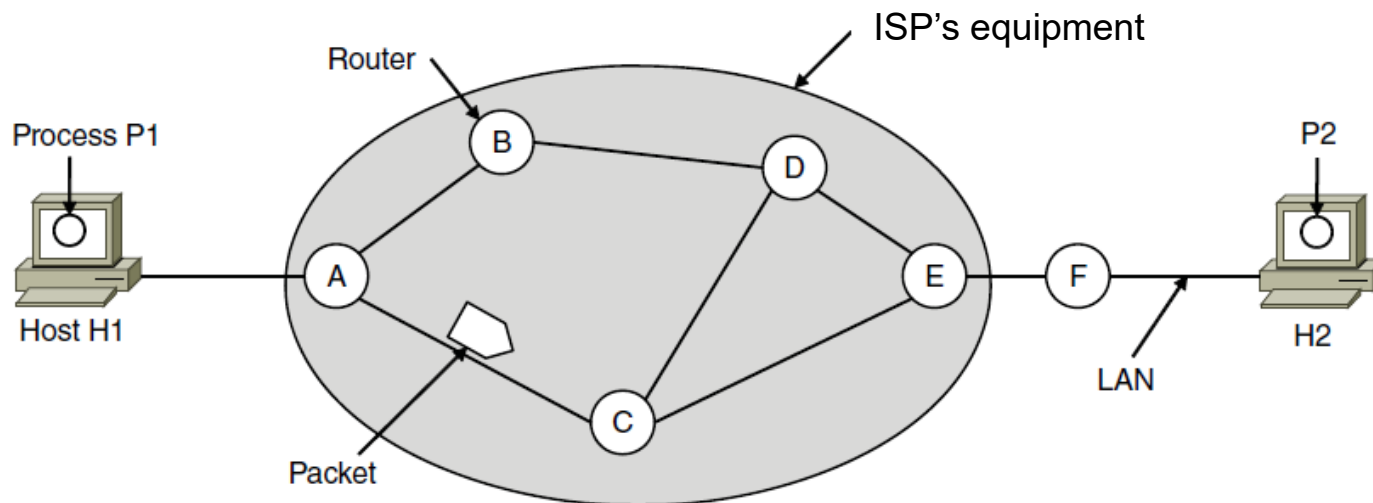
- Internet is a collection of interconnected networks that communicate using the Internet Protocol (IP)
- IP provides a **best-effort** service to **route datagrams** from source host to destination host
- The source and destination hosts may be on different networks

# Network Layer on the Internet (3)



# Store-and-Forward Packet Switching

- **Hosts** generate packets and inject into the network
- **Routers** route packets to their destination
  - Treat packets as messages, **receive/store** them and **forward** them based on the address



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# Network Layer Services

- Provided to the transport layer, independent of router technologies
- Transport layer should be shielded from the number, type and topology of routers
- Network addressing should use a uniform numbering scheme as a network identifier